Sean Carter

Mobile: (303) 434-2272, Email: sean.dan.carter@gmail.com, Website: https://collisteru.github.io
LinkedIn: linkedin.com/in/sean-carter-software-engineer

Experienced, articulate developer and researcher specializing in Artificial Intelligence (AI) implementation, research, analysis and data. Skilled at communicating cross-functionally within the enterprise and with clients.

Work Experience

Web Design Consultant, February 2026: Designed and implemented the sidenote system for the blog Philosophical Multicore (https://mdickens.me/2024/09/26/outlive_a_critical_review/).

Consulting AI Developer/Analyst, March, June 2025: Architected and developed a full stack, AI enabled, dynamic appointment calendaring system for a company which provides a time tracking product for law firms. This application summaries Zoom meeting transcripts, including next steps, and then inputs this into a calendar event. Developed a fully autonomous AI agent based phone answering system for a dental industry call center. Make, TypeScript, React, Axios, Zoom, tailwindCSS, Prisma, PostgreSQL, Windows 11.

AI Developer/Analyst, ToolCASE LLC, September 2023 to June 2024: Responsible for analyzing and improving Rembrandt AI system output accuracy with banking fraud transaction detection. Expanded model capabilities by representing transactions as images for processing by a Convolutional Neural Network (CNN). Created new Recurrent Neural Network (RNN) architecture for individual fraud detection. Performed factor analysis of mixed data for transaction data. AI transaction filtration, CNN use cases, Oracle, SQL, Python, SQL, C++, bash, Atlassian Suite, macOS, UNIX.

Machine Learning Researcher, ToolCASE LLC, June to September 2023: Responsible for two research projects to improve the Rembrandt AI financial fraud identification system. Trained a new Long Short Term Memory (LSTM) system to identify fraud on the basis of a time-series analysis of the past behavior of each individual banking customer. Used F-Score to quantify improvements from learning. Created new unsupervised learning techniques to better use data from non-fraudulent transactions. Made other research approach recommendations. Technical environment included Jupyter Notebook, Pytorch.

Software Engineer, Allthenticate, July to September 2022: Responsible for enhancement of an enterprise wide authentication system for a cybersecurity consulting and product firm. Integrated SSH public keys across bluetooth connections, the desktop applications, and the frontend. Upgraded an application with an internal API to retrieve user's public keys from the company database. Developed database queries to assign user public keys to unique emails. Encryption and cybersecurity technologies, Vue.js**,** Electron, Javascript, C++, Linux, SQL, embedded systems.

Leadership and Projects

Founding Board Member, Rocky Mountain AI Interest Group (RMAIIG) and ACX Group, Boulder, January 2023 to Present: Responsible for ensuring the meetings are set up and run effectively. Coordinated schedule for meetings among the presenters and members. Resolved logistical issues in a team. Became a prominent and known AI Specialist in the Boulder/Denver AI community.

Software Developer, First Place Winner, BlasterHacks, 2024: Led the concept, background and design, and coded with a team to develop a web based application, *PhonoPro*, which translates and trains users to learn the International Phonetic Alphabet, including the sound of each letter. It is the first of its kind. This project won first place in the education category. Coded in Javascript.

AI-Specific Skills, Development Methodologies, Technologies

Machine Learning: Developed multiple RNN, LSTM, CNN, and Transformers for novel projects. Used tools such as PyTorch, Pandas, NumPy and scikit-learn.

AI Platforms: OpenAI API, CrewAI, Rembrandt AI, language agents, and Large Language Models (LLM).

Development Methodologies: Agile Spring model used at ToolCASE LLC, scrum, waterfall.

Programming Languages and Libraries: Python, Java, SQL, C++, C, KaTeX, PrismJS, Prisma, JavaScript, Replit, JSFiddle, React, Haskell, VSCode.

Development Environments: Jupyter Notebook, WSL, Oracle, Netlify, Gatsby, Linux, Bash, RegEx, PC, Macintosh.

Education

University of Colorado, Boulder: Bachelor of Science in Computer Science (BSCS), with a Minor in Applied Mathematics, 2025. Previous undergraduate study at the University of California, Santa Barbara. Coursework includes Introduction to Natural Language Processing, Machine Learning, Introduction to AI, Computational Methods, Markov Methods, Mathematical Foundations of Probability Theory.

Sean Carter

References

Jacob Gore PhD, CTO, ToolCASE, LLC., 303-917-4165. Dr. Gore was my supervisor, whom I made AI research recommendations and provided status updates.

Max Farfel, Software Engineer, ToolCASE, LLC., 303-330-3854. I worked closely with Max as my peer at ToolCASE.

Jim Dykes, Teaching Professor in Computer Science, CU Boulder, 303-709-4254, Email preferred: james.dykes@colorado.edu. Jim was one of my advisors for my AI research senior thesis project.

Bernie Conrad, Software Engineer, Notion, Inc., 925-490-9757. Bernie was my immediate supervisor at Allthenticate, LLC.

Chad Spensky PhD, CEO and Chief Cybersecurity Developer, Allthenticate, LLC., 740-632-6257. Dr. Spensky is a researcher at MIT, a specialist in cybersecurity, and owner of Allthenticate. He was my hiring manager and reviewed my performance.

Michael Dickens, web@mdickens.me. A client for freelance projects.